

Sealed Experiment Record

Sealed 2026-10-07 · Signed by the K-Veritas server
Session 77ef80db · Data hash 6ff2...e943

Abstract

This record seals 1 run (kv_run.sh) executed on Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz between 2026-10-07 03:24 and 03:40 UTC. Reported results: final_test_acc 94.26, best_test_acc 94.31. 1 invocation in total, 0 failed. Produced with K-Veritas [1].

What this record proves

These numbers came from the sealed code at this time, unchanged since. Whether the experiment is sound is for the reader to judge.

Verify: kveritas verify <this file> or kveritas.org/verify

1 Setup

Item	Value
Session	77ef80db-844f-41f9-9ff9-6c9f4f95e83c
Initialized	2026-10-07T03:24:52Z
Machine	167ec0d6dc9a8170
Operating system	linux / amd64
CPU	Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores
Memory	31.0 GB
GPU	NVIDIA GeForce RTX 5060 Ti, 16311 MiB
Environment, run 1	44475475cf25ec56f4813ad97dde08c5e3dde73c35c164aed3de7e4221e3082f
Disclosure	open
Source code, run 1	f5242695a806564fd2325d0d39d6bc82ba8e1147febd8e3bc756cf91d96a2ad3

2 Runs

#	Command	Duration	Exit	Anchored
1	bash kv_run.sh	16 minutes 0 seconds	0	at run end

1 invocation in total: 1 kept in this record, 0 failed, 0 interrupted. Every invocation is recorded by the server, including the ones not kept.

3 Results

Run 1: kv_run.sh

Metric	Final value	Step	Source
train_loss	0.007	49 (50 readings)	explicit
train_acc	99.864	49 (50 readings)	explicit
test_loss	0.214	49 (50 readings)	explicit
test_acc	94.26	49 (50 readings)	explicit

Inline claim	Value	Phase	Output line
final_test_acc	94.26		433
best_test_acc	94.31		434

4 Execution evidence

Coherence

Coherent (0.41): the telemetry channels move together, as one process.

Telemetry

Run	Samples	CPU time	Peak memory	Peak GPU
1	7200	977.3 s	4.85 GB	96%

5 Provenance

Run 1: kv_run.sh

Disclosure open. 22 tracked files, 4 snapshots. Every snapshot is bound into the signature.

#	Snapshot	Time	Root	Changed
0	run_start	03:24:53	24e08795c32b9ef4	22
3	run_end	03:40:53	24e08795c32b9ef4	0

2 snapshots with no file change omitted here; full timeline in Appendix B.

Activity: 29 files read, 1 files written, 202 subprocesses. Full list in Appendix B.

6 Verification

Field	Value
Signer	Signed by the K-Veritas server
Signed at	2026-10-07T03:40:54.725282+00:00
Algorithm	RSA-PSS-SHA256, 4096-bit key
Key fingerprint	eeac81653cf44ac029c4069cf2a06d9f
Data hash	6ff23f44933eb5329ba020d13f9f7d4026815c3ce76236851c3919215a7fe943
Code bundle hash	7ad6960bb0d6a8cf891fda92051cf957898d04a20f2e0fb8fedf1583aefd6b14

Run anchors: 1 invocations were anchored at the server when they ended.

Checks a verifier runs

1. The stored canonical data hashes to the data hash above, and the signature verifies over data_hash:nonce:signed_at.
2. The signing key is the K-Veritas server key, not one supplied by the author.
3. Nothing was appended after the seal, and these pages hash to the value bound in the seal.
4. Each run matches the digest the server received when it ended.
5. A supplied code bundle hashes to the code bundle hash above.

References

[1] M. K. Keita and C. Homan. Computer Science Conferences Should Require Nonrepudiable Experimental Results. NeurIPS 2026 Position Paper Track. arXiv:2605.08586.

Appendix

A Full metric series

Run 1: kv_run.sh

Metric	Value	Step	Line	Source
train_loss	1.795	0	7	explicit
train_acc	34.462	0	8	explicit
test_loss	1.434	0	10	explicit
test_acc	47.1	0	11	explicit
train_loss	1.302	1	16	explicit
train_acc	52.254	1	17	explicit
test_loss	1.152	1	19	explicit
test_acc	57.38	1	20	explicit
train_loss	1.036	2	25	explicit
train_acc	62.928	2	26	explicit
test_loss	1.294	2	28	explicit
test_acc	57.64	2	29	explicit
train_loss	0.876	3	34	explicit
train_acc	69.296	3	35	explicit
test_loss	1.029	3	37	explicit
test_acc	64.29	3	38	explicit
train_loss	0.736	4	43	explicit
train_acc	74.27	4	44	explicit
test_loss	0.752	4	46	explicit
test_acc	73.97	4	47	explicit
train_loss	0.641	5	52	explicit
train_acc	77.892	5	53	explicit
test_loss	0.662	5	55	explicit
test_acc	77.87	5	56	explicit
train_loss	0.583	6	61	explicit
train_acc	79.844	6	62	explicit
test_loss	0.666	6	64	explicit
test_acc	77.65	6	65	explicit
train_loss	0.541	7	69	explicit
train_acc	81.288	7	70	explicit
test_loss	0.717	7	72	explicit
test_acc	76.91	7	73	explicit
train_loss	0.505	8	77	explicit
train_acc	82.598	8	78	explicit
test_loss	0.573	8	80	explicit
test_acc	79.69	8	81	explicit
train_loss	0.486	9	86	explicit
train_acc	83.194	9	87	explicit
test_loss	0.712	9	89	explicit
test_acc	77.05	9	90	explicit
train_loss	0.457	10	94	explicit
train_acc	84.318	10	95	explicit
test_loss	0.466	10	97	explicit
test_acc	84.15	10	98	explicit

Metric	Value	Step	Line	Source
train_loss	0.44	11	103	explicit
train_acc	85.034	11	104	explicit
test_loss	0.544	11	106	explicit
test_acc	81.7	11	107	explicit
train_loss	0.416	12	111	explicit
train_acc	85.616	12	112	explicit
test_loss	0.534	12	114	explicit
test_acc	81.79	12	115	explicit
train_loss	0.402	13	119	explicit
train_acc	86.258	13	120	explicit
test_loss	0.739	13	122	explicit
test_acc	78	13	123	explicit
train_loss	0.38	14	127	explicit
train_acc	87.038	14	128	explicit
test_loss	0.607	14	130	explicit
test_acc	79.41	14	131	explicit
train_loss	0.369	15	135	explicit
train_acc	87.468	15	136	explicit
test_loss	0.57	15	138	explicit
test_acc	80.32	15	139	explicit
train_loss	0.36	16	143	explicit
train_acc	87.632	16	144	explicit
test_loss	0.614	16	146	explicit
test_acc	80.72	16	147	explicit
train_loss	0.342	17	151	explicit
train_acc	88.31	17	152	explicit
test_loss	0.559	17	154	explicit
test_acc	81.85	17	155	explicit
train_loss	0.328	18	159	explicit
train_acc	88.804	18	160	explicit
test_loss	0.497	18	162	explicit
test_acc	83.85	18	163	explicit
train_loss	0.321	19	167	explicit
train_acc	88.954	19	168	explicit
test_loss	0.46	19	170	explicit
test_acc	84.54	19	171	explicit
train_loss	0.311	20	176	explicit
train_acc	89.352	20	177	explicit
test_loss	0.443	20	179	explicit
test_acc	85.26	20	180	explicit
train_loss	0.291	21	185	explicit
train_acc	90.072	21	186	explicit
test_loss	0.512	21	188	explicit
test_acc	83.32	21	189	explicit
train_loss	0.277	22	193	explicit
train_acc	90.496	22	194	explicit
test_loss	0.44	22	196	explicit
test_acc	85.3	22	197	explicit
train_loss	0.272	23	202	explicit
train_acc	90.628	23	203	explicit

Metric	Value	Step	Line	Source
test_loss	0.467	23	205	explicit
test_acc	84.54	23	206	explicit
train_loss	0.26	24	210	explicit
train_acc	91.084	24	211	explicit
test_loss	0.507	24	213	explicit
test_acc	83.9	24	214	explicit
train_loss	0.246	25	218	explicit
train_acc	91.564	25	219	explicit
test_loss	0.351	25	221	explicit
test_acc	88.32	25	222	explicit
train_loss	0.233	26	227	explicit
train_acc	91.922	26	228	explicit
test_loss	0.354	26	230	explicit
test_acc	88.03	26	231	explicit
train_loss	0.217	27	235	explicit
train_acc	92.482	27	236	explicit
test_loss	0.36	27	238	explicit
test_acc	88.56	27	239	explicit
train_loss	0.205	28	244	explicit
train_acc	92.87	28	245	explicit
test_loss	0.564	28	247	explicit
test_acc	83.47	28	248	explicit
train_loss	0.196	29	252	explicit
train_acc	93.194	29	253	explicit
test_loss	0.338	29	255	explicit
test_acc	88.61	29	256	explicit
train_loss	0.178	30	261	explicit
train_acc	93.856	30	262	explicit
test_loss	0.297	30	264	explicit
test_acc	90.45	30	265	explicit
train_loss	0.16	31	270	explicit
train_acc	94.532	31	271	explicit
test_loss	0.337	31	273	explicit
test_acc	89.35	31	274	explicit
train_loss	0.152	32	278	explicit
train_acc	94.834	32	279	explicit
test_loss	0.29	32	281	explicit
test_acc	90.6	32	282	explicit
train_loss	0.132	33	287	explicit
train_acc	95.518	33	288	explicit
test_loss	0.312	33	290	explicit
test_acc	90.29	33	291	explicit
train_loss	0.118	34	295	explicit
train_acc	95.974	34	296	explicit
test_loss	0.337	34	298	explicit
test_acc	89.66	34	299	explicit
train_loss	0.101	35	303	explicit
train_acc	96.542	35	304	explicit
test_loss	0.281	35	306	explicit
test_acc	91.31	35	307	explicit

Metric	Value	Step	Line	Source
train_loss	0.088	36	312	explicit
train_acc	97.022	36	313	explicit
test_loss	0.259	36	315	explicit
test_acc	92.08	36	316	explicit
train_loss	0.071	37	321	explicit
train_acc	97.618	37	322	explicit
test_loss	0.255	37	324	explicit
test_acc	92.3	37	325	explicit
train_loss	0.058	38	330	explicit
train_acc	98.152	38	331	explicit
test_loss	0.253	38	333	explicit
test_acc	92.4	38	334	explicit
train_loss	0.049	39	339	explicit
train_acc	98.438	39	340	explicit
test_loss	0.239	39	342	explicit
test_acc	92.98	39	343	explicit
train_loss	0.037	40	348	explicit
train_acc	98.788	40	349	explicit
test_loss	0.236	40	351	explicit
test_acc	93.1	40	352	explicit
train_loss	0.027	41	357	explicit
train_acc	99.148	41	358	explicit
test_loss	0.234	41	360	explicit
test_acc	93.47	41	361	explicit
train_loss	0.019	42	366	explicit
train_acc	99.412	42	367	explicit
test_loss	0.223	42	369	explicit
test_acc	93.77	42	370	explicit
train_loss	0.016	43	375	explicit
train_acc	99.53	43	376	explicit
test_loss	0.222	43	378	explicit
test_acc	93.93	43	379	explicit
train_loss	0.011	44	384	explicit
train_acc	99.762	44	385	explicit
test_loss	0.218	44	387	explicit
test_acc	94.1	44	388	explicit
train_loss	0.01	45	393	explicit
train_acc	99.788	45	394	explicit
test_loss	0.219	45	396	explicit
test_acc	94.07	45	397	explicit
train_loss	0.009	46	401	explicit
train_acc	99.828	46	402	explicit
test_loss	0.214	46	404	explicit
test_acc	94.3	46	405	explicit
train_loss	0.007	47	410	explicit
train_acc	99.88	47	411	explicit
test_loss	0.212	47	413	explicit
test_acc	94.31	47	414	explicit
train_loss	0.007	48	419	explicit
train_acc	99.844	48	420	explicit

Metric	Value	Step	Line	Source
test_loss	0.212	48	422	explicit
test_acc	94.29	48	423	explicit
train_loss	0.007	49	427	explicit
train_acc	99.864	49	428	explicit
test_loss	0.214	49	430	explicit
test_acc	94.26	49	431	explicit

Phase	Line	Time	CPU time	Memory	GPU
train	1	03:24:53	0.4 s	0.41 GB	
summary	432	03:40:53	0.0 s	0.00 GB	

B Provenance timeline and activity map

Run 1: kv_run.sh

#	Snapshot	Time	Root	Changed
0	run_start	03:24:53	24e08795c32b9ef4	22
1	phase train	03:24:53	24e08795c32b9ef4	0
2	phase summary	03:40:53	24e08795c32b9ef4	0
3	run_end	03:40:53	24e08795c32b9ef4	0

Event	Path or command	Hash
read	kv_run.sh	
read	main.py	
read	models/densenet.py	
read	models/dla.py	
read	models/dpn.py	
read	models/lenet.py	
read	models/mobilenet.py	
read	models/mobilenetv2.py	
read	models/preact_resnet.py	
read	models/senet.py	
read	models/shufflenet.py	
read	models/shufflenetv2.py	
read	models/vgg.py	
read	models/__init__.py	
read	models/pnasnet.py	
read	models/googlenet.py	
read	models/resnet.py	
read	models/resnext.py	
read	models/efficientnet.py	
read	models/regnet.py	
read	models/dla_simple.py	
read	utils.py	
read	data/cifar-10-batches-py/data_batch_1	
read	data/cifar-10-batches-py/data_batch_2	
read	data/cifar-10-batches-py/data_batch_3	
read	data/cifar-10-batches-py/data_batch_4	
read	data/cifar-10-batches-py/data_batch_5	
read	data/cifar-10-batches-py/test_batch	

[illegible]

[illegible]

[illegible]

[illegible]

Event	Path or command	Hash
process	../env/bin/python -u main.py --lr 0.1	
process	../env/bin/python -u main.py --lr 0.1	
process	../env/bin/python -u main.py --lr 0.1	
process	../env/bin/python -u main.py --lr 0.1	

C Run history

Invocation 1, OK, 16 minutes 0 seconds, 434 output lines, started 2026-10-07T03:24:52

metric_hash 465e3f8141f002c969c158237261f2ffa9202332900352fb043eb8b7b02e5258
run_digest 3dd6338ef40c62bb9c11881f9aad0e26cdeb245f154ef01affd7fd041b5c340c
anchored at run end

D Raw seal fields and hashes

Seal

data_hash 6ff23f44933eb5329ba020d13f9f7d4026815c3ce76236851c3919215a7fe943
signed_message_hash f7c89998405490cb179f11f379f3c0956146bcb497f645fcade04918583bc202
nonce dd131a40bb4c582937eeb663d157a578
signed_at 2026-10-07T03:40:54.725282+00:00
sealed_at 2026-10-07T03:40:54.785861509Z
checkout_bundle 7ad6960bb0d6a8cf891fda92051cf957898d04a20f2e0fb8fedf1583aefd6b14

Signature (base64)

TW4lAVYQPliLeckLrglX4m+9IWCA6IX9rQGhNNwSwcCp9p6payerWlzIxFS3izAn01StDZe6bKQPPQ41CC3G5v/v
ON1u66E/rLzX4/CUD0BhhQ0bmEQ+u+bt3w1HW6j+r0RPAi4oIjBQuRJgWTRG3kp6EuliOKzDf2D/wtAgRcj2rDn
PypBC86/gJ3uEjJ40MYRCldUm9u0RdrUdYqVL7TsdaxuMaCRU/13dfZPgviz1TvAxGDSCa7DI5ZaTfPGR7Yik400
meVsusXPLjPPxi9SxxNUN4ugoAyP9a1TpR8fLT0uMwQ+PmliHs0YBB0Py+j17wXw9ap8+5/OgtzYsucjkVeBIkCi
PjRiESlVRyKyK9//V1k+WAI4mVWwafFURiAUR5L9p5X1knxxN7FGayNtQwgDvtrLI0bTk5S6k56KZDX0oy052Qa0
R2Dtt19t0ZAuFjTDlpBBJu/oTaHeJw051eDr2VZC+FZWZo2QPji1+V9sGB3KtCXr4+2GPzjcpK0lVDLRNfTabXZZ
Jp4BUYBX3P7z+S9peHTxtL8YpDZpHhHZOGIF0tc6GLQneq2cqMq9Iwfn883QSc1hrCW9Y8ZvjHLeHCfdz8ezle3N
lOYfzji/+xisHu4xHfB13cIIGPFf3x6P33hsiLby2bFtzAejLbiqaQA6xxc+WNRfoU=

Public key

-----BEGIN PUBLIC KEY-----
MIICIjANBgkqhkiG9w0BAQEFAAOCAg8AMIICGKCAgEAE1QL3YyMgeShx8IhTC7jb
sM0w2BnRXgsBn3+SMXKjtwHATqdekmlPlcjayJxD5fcPrOPPCqp/LoUFz5CUZzRi
6+8FTv70dHuIz5A4Lat4vcgwP+iWscPc3HzpmjpsKbTm0emqgp6RB6GnnPpxYa5X
7qu95EfEisX/GfoS9ToxtqPBwCvWSPBX8efec4TC58m96Fsp4mC8KicEadbIbHN
av0JjdTJ//CMOWZSIbdzE6qvEbNFCqDB5tNU7oHfYvBeNLrvPU9TAqdbtozdHq+D
AhfTRuJNgWEkjWd5l0Vhi4qlkweS0EMTKT3ye9IL5xEaJn4bku4eKqes7VSZt3o
XbUcmr2H5mKcKsvt6bXmX6ajAmgi900Cm9QJlaQkJM6z2iITnyvDMgueJtFxQpC9
HNBPGpZ242NH8JgE6CNF7ToL9syasS67u7UMgIcNgeNHNCy9bLgPU5annzP9gN2l
6HEMldkXNdKQRb1rDwyEJf3N/50YHnXXPmuf5sb8R/pxWNqtF04R5UGVV92qd2Mi
1GN7lE3mbQRbPsLL/JApDXMNzNikaFFBsQK7iKxr7vIaTeZckFZWfThcYNT7Hasg
+eE4bf/olCHBCf1VIDCkDxohehdZ0WRRcgM9iK3I/0zZdaBsgdc8WuJRMb8LPP9v
8Jap/1PwjNcNku5cdw3gHLCCAwEAAQ==
-----END PUBLIC KEY-----

Run 1 outputs

stdout a350a59c36c02ae4ed4061df1cafa11836802270ceeeaf2609eed39897723dc6
stderr e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855
metrics 465e3f8141f002c969c158237261f2ffa9202332900352fb043eb8b7b02e5258
digest 3dd6338ef40c62bb9c11881f9aad0e26cdeb245f154ef01affd7fd041b5c340c
kv_run.sh (unchanged)
pre 5653a85614d28fac9f772c5650562e420fc9ba2bd5b72f828fcff4a8c9d850a2
post 5653a85614d28fac9f772c5650562e420fc9ba2bd5b72f828fcff4a8c9d850a2

E BibTeX for [1]

```
@misc{keita2026computerscienceconferencesrequire,  
  title={Computer Science Conferences Should Require Nonrepudiable Experimental Results},  
  author={Mamadou K. Keita and Christopher Homan},  
  year={2026},  
  eprint={2605.08586},  
  archivePrefix={arXiv},  
  primaryClass={cs.CR},  
  url={https://arxiv.org/abs/2605.08586},  
}
```